
Take-Home Task (2 days)

Full-Stack Lead Management App

Problem Statement

Build a simple internal web application that allows a team to:

- Capture new leads
- Track follow-ups
- Add notes to conversations
- View a history of interactions with each lead

This simulates a **lightweight CRM system** used by small teams to track potential customers.

The goal is to evaluate your ability to design a **clean full-stack application**, including:

- database design
- API design
- UI implementation
- validation and structure

Recommended Tech Stack

You may use any modern stack you are comfortable with.

Examples:

Frontend

- React
- Next.js
- Vue

Backend / Database

You may choose either:

Option A (Recommended)

- Supabase (Auth + Database)

Option B

- Node.js / Express / FastAPI backend
- PostgreSQL / SQLite

If using Supabase, you may use **Supabase Auth** for login.

Scope & Requirements

1. Authentication

Implement a simple login system.

Requirements

- Email + password login
- Single team role (no complex permissions required)

Using **Supabase Auth** is recommended but not required.

After login, users should be able to access the application dashboard.

2. Lead Management (CRUD)

Users should be able to **create, view, and manage leads**.

Lead Data Model

Table: leads

Column	Type
id	uuid
name	text (required)
phone	text (required, unique)
city	text
source	text
status	text
created_at	timestamp

Field Definitions

name
Name of the lead or contact person.

phone
Phone number of the lead.
Must be **unique**.

city
Optional city of the lead.

source

Where the lead came from. Example values:

- website
- whatsapp
- referral
- ads

status

Lead status:

- new
- follow_up
- closed

created_at

Timestamp when the lead was created.

3. Required Screens

Leads List Page

This is the main dashboard.

The page should display a list of leads.

Each row should show:

- name
 - phone
 - city
 - source
 - status
 - created date
-

Search

Users should be able to search leads by:

- phone
 - name
-

Filters

Allow filtering by:

- status
- source

Example:

Status: follow_up
Source: whatsapp

Create Lead Page

Users should be able to create a new lead.

Required fields:

- name

- phone

Optional fields:

- city
- source
- status

Example form submission:

```
{
  "name": "Tharun Sharma",
  "phone": "+91 99555 52671",
  "city": "Hyderabad",
  "source": "website",
  "status": "new"
}
```

Lead Detail Page

Clicking a lead should open a **Lead Detail page**.

The page should show:

- Lead information
 - Notes timeline
 - Form to add a new note
-

4. Notes Timeline

Each lead should support **multiple notes**.

Notes represent interactions or updates about the lead.

Examples:

- "Customer asked for pricing"
 - "Follow-up call scheduled for tomorrow"
-

Table: lead_notes

Column	Type
id	uuid
lead_id	uuid
note_text	text
created_at	timestamp

Behavior

Notes should appear in **reverse chronological order**.

Example:

10:30 AM — Sent pricing information
09:15 AM — Customer asked for product demo
Yesterday — Initial contact

Users should be able to **add a new note** from the lead detail page.

5. Automation–Ready API Endpoint

The application should expose simple API endpoints that allow external systems to create leads and notes.

This simulates **automation workflows** where other systems send data to the CRM.

Create Lead API

POST /api/leads

Example request:

```
{
  "name": "Priya Kumar",
  "phone": "+91 99555 52672",
  "city": "Warangal",
  "source": "whatsapp",
  "status": "new"
}
```

Add Note API

POST /api/leads/:id/notes

Example request:

```
{
  "note_text": "Customer requested pricing details"
}
```

API Security

These endpoints should be protected using a simple header:

x-api-key: my_secret_key

The server should verify this header before processing requests.

Store the API key in an **environment variable**.

Bonus Features (Optional)

These features are **not required but strongly encouraged**.

Duplicate Phone Handling

If a user attempts to create a lead with a phone number that already exists:

Instead of creating a duplicate record:

- show a message such as **"Lead already exists"**
- redirect to the existing lead page.

Export Leads to CSV

Provide a button that exports leads as a CSV file.

Example:

```
name,phone,city,source,status
Tharun Sharma,+919955552671,Hyderabad,website,new
Priya Kumar,+919955552672,Warangal,referral,follow_up
```

Deliverables

Please submit:

1. GitHub Repository

Your repository should contain the complete project.

2. README

Include:

- setup instructions
- environment variables
- database setup
- how to run locally

Example:

```
npm install
npm run dev
```

3. API Testing Examples

Provide either:

- curl examples
- or
- Postman collection

Example:

```
curl -X POST http://localhost:3000/api/leads \
-H "x-api-key: my_secret_key" \
-H "Content-Type: application/json" \
-d '{"name": "Tharun", "phone": "+919955552672" }'
```

Evaluation Criteria (Guidance)

We primarily evaluate:

- Clean **database schema**
- Well-structured **frontend UI**
- Correct **CRUD functionality**
- Clear **API implementation**

- Basic **validation and error handling**
- Overall **code organization**

Bonus points for:

- Good UI/UX
 - Handling duplicate leads
 - CSV export
 - Clean component structure
 - Good documentation
-

Expected Outcome

A working full-stack web application where a team can:

- log in
 - manage leads
 - add notes
 - track follow-ups
 - receive leads through a simple API
-